Cryptocurrency pair spreads: a screening-artifact audit and the marginal survivor

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Abstract

We set out to build a profitable statistical arbitrage strategy on hourly cryptocurrency pair spreads. It did not work, and this paper is mostly about why. Two screens that appear to find strong, tradable structure are statistical artifacts. A Kalman dynamic hedge ratio appears to produce stationary, mean reverting spreads for nearly the entire liquid universe, but it runs a stationarity test on a filter's own innovations, which are white by construction. Placebos that cannot be cointegrated pass the same screen at the same nearly 100% rate, while a clean test finds genuine out of sample cointegration in only a few percent of pairs. A rolling z score appears to generate a large reversion edge, but it reverts mechanically even on a pure random walk. Both artifacts hold at hourly, four hour, and daily frequencies. Once we benchmark against placebos, the structure that survives is either priced too fast to trade or too marginal to deploy. Order flow information in the L2 book decays within seconds. Reversion speed ranks pairs (Spearman $\rho = 0.46$), but the ranking does not turn into profit. The one market neutral mean reversion book that clears honest accounting posts a HAC and venue adjusted monthly Sharpe near 1.0,[1] and only without a stop and only as a multiweek hold. That 1.0 is an exploratory, post hoc estimate that was never preregistered, and our preregistered test gives a confidence interval spanning zero, so we claim no significant tradable edge. We read this as evidence that the liquid core of crypto pair spreads is efficient. The methodological lesson is simpler: do not trust a cointegration or z score screen until you have shown it fails on a matched placebo.

1 Introduction

The goal at the outset was a profitable hourly pairs strategy on crypto. We did not get one, and the useful finding is why: the two screens that made the strategy look best are screen artifacts, and whatever structure is genuinely present is either priced within seconds or too thin to trade once costs and selection are counted.

The project began in the pairs trading tradition of Gatev, Goetzmann, and Rouwenhorst [12]: find two assets whose prices hold a stable relationship, trade the spread when it deviates, and collect the reversion. Cryptocurrency is a natural place to look, with a continuously traded market and a rapidly changing set of listed coins. Our proposal was noncommittal about profit from the start. It framed the goal as quantifying whether stable mean-reverting relationships exist in crypto, "as a way of gaining a deeper understanding of market efficiency and information flow." This paper answers that question, and the answer is mostly negative.

We claim three contributions, each bounded against prior work as cited. The first rests on empirical demonstration plus a conjectural transfer of the Eroğlu, Miller, and Yiğit mechanism to

[1] All Sharpe figures in this paper are unit notional and capacity unbounded; impact is not modeled. See Limitations.

the frozen-filter innovation sequence; a formal proof at that level is open. With that caveat: first, two standard pair screening tools manufacture the appearance of structure on data that cannot carry it. A Kalman dynamic hedge cointegration test passes random walk, phase-randomized, and block-shuffled placebos at the same near-100% rate it passes real pairs, and a positive control of constructed cointegrated pairs passes at the same rate as the random-walk negative: at the critical values used in this pipeline, the screen does not separate real pairs from matched placebos. A rolling z score reverts mechanically on pure random walks, producing per-event reversion of $+0.98z$ to $+1.61z$ across hourly, four-hour, and daily frequencies and a Sharpe near 2.2 on random-pair placebos against 4.75 on the actual data. The underlying mechanisms are known: a well specified Kalman filter has white steady state innovations [13], rolling mean demeaning induces negative autocorrelation [30, 5], and Eroğlu, Miller, and Yiğit [10] show that time-varying-parameter Kalman models are vulnerable to spurious regression because the integrated error is transferred to the estimated state equation. Our contribution is the screen-level placebo audit: for the Kalman ADF screen, on the Kalman dynamic hedge plus ADF pipeline that circulates in pairs trading practice ([28, 8, 26]), with a positive control showing the screen cannot separate constructed cointegrated pairs from random walks; and for the rolling z rule, the random-real-pairs bootstrap of Gatev, Goetzmann, and Rouwenhorst [12] is the closest prior precedent, extended here to synthetic random-walk and phase-randomized nulls with per-event reversion magnitudes reported in z-units. Second, extending Schnaubelt's [24] large-scale RL placement study on crypto L2, we run a specification-style test on a four-major basket (BTC, ETH, SOL, AVAX) with 29,952 parent orders on event-level Binance L2 data in 2024. Aggressive crossing is cheapest across all four symbols, the signal correlates +0.03 with the per-order advantage of posting over crossing, and a perfect foresight oracle bounds achievable savings at 1.4 bps: the opportunity exists but no contemporaneous feature realizes it. Third, the candidate market-neutral reversion strategy that emerges from the surviving structure clears honest accounting at a venue- and HAC adjusted monthly Sharpe near 1.0, only without a stop, and a preregistered held-out backtest of it inflates above the in-sample number on uncostable microcap moves; our one surviving result ends up cutting against itself.

Across all three the lens is the same: the screen that produced the candidate is the part to stress, not the candidate itself. The backtest-overfitting literature [3] stress tests the strategy you finally pick; the screen you used to build the spread needs the same scrutiny.

To restate the scope. The Kalman screen mechanism is from Eroğlu, Miller, and Yiğit [10] and our extension is the screen-level placebo audit and the positive control. The L2 execution test extends Schnaubelt [24] with an oracle bound and a null result on a smaller basket. The survivorship analysis combines point-in-time crypto cross-sections (Ammann, Burdorf, Liebi, and Stöckl [1]; Liu, Tsyvinski, and Wu [16]) with a leg-level break-aware stop (Lu et al. [20]) and asks the joint question for crypto pair spreads. None of the three is unconditional novelty; each is a deliberate addition on top of cited prior work.

2 Related work

Three bodies of prior work set up this paper. The two screens we audit come from the cointegration branch of the pairs trading literature. The order flow tools we apply to crypto L2 data are standard market microstructure. And the crypto market efficiency literature is where our headline negative result has to be read.

2.1 Pairs trading and dynamic cointegration

The cointegration branch of pairs trading descends from Engle and Granger [9], with its practitioner facing version codified in Vidyamurthy [28]: fit a spread, test the residual for stationarity, trade deviations of the residual. The two screens we audit in Section 4 are standard implementations of that pipeline. The state space variant comes from Elliott, van der Hoek, and Malcolm [8], who model the spread as a discrete time Ornstein-Uhlenbeck process observed in Gaussian noise. Triantafyllopoulos and Montana [26] extend it to time varying parameters with online estimation, which is the Kalman dynamic hedge we run. The distance based alternative of Gatev, Goetzmann, and Rouwenhorst [12] is the standard benchmark. Harvey's classical state space text [13] states the steady state whitening property we exploit. The closest econometric prior is Eroğlu, Miller, and Yiğit [10], who show that time-varying-parameter Kalman cointegration models suffer from spurious regression because the integrated error is transferred to the estimated state equation; they propose corrections at the cointegrating-vector level. Our paper applies the consequence of that finding at the screen level: the specific Kalman-plus-ADF pipeline used as a tradability filter cannot discriminate cointegrated pairs from random walks, and we document this empirically with matched placebos and a positive control.

2.2 Microstructure and order flow

The information content of order flow is the central object of classical microstructure, beginning with Kyle [18] on the linear price impact of informed trading. Hasbrouck [14] introduces the permanent versus transient decomposition of trade impact, which we use to interpret the seconds-scale half-life we observe. The event-time order flow imbalance of Cont, Kukanov, and Stoikov [6] predicts sub-second mid moves nearly linearly on equities; we replicate this on Binance L2 in 2024 for BTC, ETH, SOL, and AVAX. We use the size-weighted microprice of Stoikov [25] as our regression target. The toxicity program of Easley, López de Prado, and O'Hara [7] is the natural place to look for a tradable order flow signal at longer horizons, but Andersen and Bondarenko [2] show its predictive content is largely mechanical and its sign is sensitive to trade classification. Our crypto VPIN measurement replicates their critique.

2.3 Cryptocurrency market efficiency

The crypto efficiency literature has converged on a picture our results are consistent with: the most-traded coins on the most-traded venues look weak-form efficient, while thinner segments do not. Urquhart's [27] original Bitcoin tests rejected EMH on early daily data but already noted the second half of the sample looked tighter. Wei [29] shows on 456 coins that return predictability and Hurst persistence collapse with liquidity, and Brauneis and Mestel [4] replicate the link across 73 coins and eight tests. Makarov and Schoar [21] show that persistent cross-exchange price gaps survive transaction costs and are best explained by capital-control and credit frictions, which mirrors why our Coinbase adjusted Sharpe sits below the Binance.US figure. The closest prior on crypto pairs is Fil and Kristoufek [11], who find that most strategies underperform their benchmarks once costs are realistic; our honest accounting Sharpe of $\sim$1.0 is consistent with their picture.

2.4 Machine learning approaches

Krauss, Do, and Huck [15] apply deep neural networks, gradient boosted trees, and random forests to S&P 500 statistical arbitrage and find ensemble lifts that erode sharply after costs. Our XGBoost / LSTM / transformer comparison reproduces the same pattern on crypto pairs: a calibrated spread

z score carries roughly 79% of the P&L of a four architecture stack, in line with the multiple testing cautions of López de Prado [19]. The deep-learning literature on limit order book data predicts at sub-second horizons, which is shorter than our hourly bars and consistent with our finding that L2 order flow information decays within 30 seconds and does not survive to a tradable horizon.

2.5 Position relative to prior work

Each of this paper's three contributions is bounded against prior art. We list what we add on top of the cited literature, not what is unconditionally new. First, the screen-level placebo audit of the Kalman dynamic hedge plus ADF cointegration pipeline. The mechanism that drives our finding, a quasi-maximum-likelihood (QMLE) fit Kalman filter absorbing integrated error into the time-varying state estimate, is documented at the cointegrating-vector level by Eroğlu, Miller, and Yiğit [10]; the screen-level empirical consequence we report (the specific Kalman-plus-ADF tradability filter passes random walk, phase-randomized, and block-shuffled placebos at the same near-100% rate as real pairs, and a positive control of constructed cointegrated pairs passes at the same rate as the random walk negative) is the contribution. The companion rolling z result is a similarly screen-level placebo audit of a known induced-autocorrelation artifact [30, 5]; the closest prior precedent is the random-real-pairs bootstrap of Gatev, Goetzmann, and Rouwenhorst [12], and our extension is the specific combination of synthetic random-walk and phase-randomized nulls applied to a rolling-window z-score screen with per-event reversion magnitudes reported in z-units. Second, building on Schnaubelt [24], who shows that L2-aware reinforcement-learning placement can beat an always-cross immediate-market baseline on cryptocurrency pairs at much larger scale (18 months, ∼3.5M order book states), we run a complementary specification-style test on a four-major basket (BTC, ETH, SOL, AVAX) with 29,952 parent orders on Binance 2024 L2 data. The contribution is narrower and twofold: a perfect-foresight oracle as an explicit upper bound, which bounds the achievable savings at 1.4 bps, and a negative finding that no simple contemporaneous L2 imbalance feature realizes that oracle gap (signal-vs-cross correlation +0.03). Third, we apply two ingredients jointly to crypto pair spreads: a point-in-time universe that respects the actual delisting dates of LUNA, UST, FTT, and LUNC (PIT and delisting-aware crypto cross-sections appear in Ammann, Burdorf, Liebi, and Stöckl [1] and Liu, Tsyvinski, and Wu [16]), and a leg-level structural-break circuit breaker that halts a pair when one leg moves more than 50% adversely (break-aware stops on equity pair spreads appear in Lu et al. [20]). The breaker preserves roughly 96% of monthly Sharpe at an 8%/year breakage rate consistent with historical crypto delisting frequency. The novel element here is the joint application of these two ingredients on crypto pair spreads; we do not claim the combination is unprecedented and each ingredient has prior art in adjacent domains as cited.

3 Data and market setting

The spread work uses hourly OHLCV for USDT spot pairs from the Binance.US API, stored locally, spanning 2021 through May 2026. The cross-exchange check pulls the same hours from Coinbase (USD quoted) for the overlapping majors. The microstructure work uses event-level L2 data from Tardis: top-25 book snapshots and the trade tape for Binance.com (global) spot, the 50 most liquid USDT symbols, over 2024. We keep these separate on purpose, and note the venue gap: the spread strategy trades Binance.US, a much thinner venue, while the order flow null is measured on the deeper global book. The spread and reversion results are Binance.US and Coinbase hourly klines. Only the microstructure results use the Tardis book.

The investable universe is the 50 most liquid USDT symbols, $\binom{50}{2} = 1{,}225$ candidate pairs, constructed as of each analysis time. A symbol is eligible only if its history begins before the analysis window and reaches its start, a point-in-time rule added after an audit found that later listings could otherwise leak into the historical universe. The local store holds currently listed symbols, so delisted coins are mostly absent. We treat this as a survivorship limitation (Section 7), not a solved problem. Costs are charged on every position change at 15 bps per-leg (10 bps fee, 5 bps slippage), about 30 bps round trip for the two-leg spread, and stated before any performance number.

A few conventions hold throughout. Inference uses Newey-West HAC standard errors [22] (Appendix A.4), because overlapping targets and month-long holds make naive errors far too small. In one case a spurious five minute reversal collapsed from $t = -22.8$ to $t = -1.4$ under HAC. Evaluation is strict walk forward, six month train and three-month test (Appendix A.7), with hedge ratios and pair selection fit on the train window only. And every positive claim is checked against a matched placebo, a random walk, phase-randomized, block-shuffled, or random-pair null built to reproduce the claim if it is mechanical (Appendix A.8). Selection across many configurations is corrected with a deflated Sharpe ratio [3] (Appendix A.5), which discounts the Sharpe ratio (a strategy's mean return divided by its return standard deviation) for the number of configurations searched. Appendix A gives self-contained definitions of every method and metric used in this paper, for readers who have not met all of them.

For the one trading result that survives (Section 6) we preregistered a single configuration: the parameters and the held-out evaluation window were written to a file and committed before the run, then evaluated once and reported whatever the result was. The lock is commit `d80cde7` ("pre-register the single-config reversion test (locked before running)"), and the one correction we made before running (`bd4909b`) is also in the commit history.

4 Two screening artifacts

Two screens drove our early results, and both are artifacts of the screen, not of the data. The first, a Kalman dynamic hedge ratio, appeared to find out-of-sample cointegration in almost every liquid pair. The second, a rolling z score, appeared to find strong mean reversion. We show that each passes a placebo it should fail, and that neither carries the information it seems to. This is the paper's main methodological contribution. The rolling z score is a standard entry signal and the Kalman screen a natural way to test a drifting hedge, so both are easy to reach for and easy to be fooled by.

4.1 The Kalman dynamic hedge cointegration screen

A static hedge ratio fits one β for the whole sample. A Kalman dynamic hedge (Appendix A.1) lets β drift as a random-walk and updates it each step from the data. We fit the process and observation noise by maximum likelihood on the training window, froze them, and rolled the filter forward on the test window. The test period innovations, the filter's one step prediction errors, became the spread, and we ran an augmented Dickey-Fuller (ADF) test on them [23] (Appendix A.2). Under this screen 100% of the top 50 pairs rejected the unit root out-of-sample at $p < 0.05$, and 96.4% at $p < 0.001$. A static OLS spread over the same window passed only about 3% of pairs. The contrast looked decisive.

The contrast measures the filter, though, and says almost nothing about the pair. A Kalman filter with nonzero process noise adapts its hedge ratio to track whatever relationship is present. Its innovations are then driven toward whiteness, exactly white only for a correctly specified filter at steady state, and approximately so here on the frozen parameter test window [13]. This is a screen-level analogue of the spurious-regression-in-Kalman-TVP mechanism documented by Eroğlu, Miller, and Yiğit [10]: when the model is estimated by quasi-maximum likelihood, the integrated component of the pair is absorbed into the time-varying state estimate, leaving stationary innovations regardless of whether the underlying processes are cointegrated. An ADF test on a nearly white series rejects the unit root almost always, whether or not the two assets are cointegrated. Running that test on a filter's own innovations is close to circular. The real evidence is in the controls below, which show that even genuinely cointegrated pairs look like random-walks under this screen.

We can state the screen-level consequence formally:

> *Conjecture (screen-level analogue, motivated by [10]).* For $I(1)$ processes y_t and x_t, cointegrated or not, fit a Kalman dynamic-hedge model $y_t = \alpha_t + \beta_t x_t + v_t$ by quasi-maximum likelihood on a training window, then forward-roll the filter with frozen parameters on a test window of length T. We conjecture that the test-window one-step innovations $\{e_t\}$ inherit no integrated component, so an ADF test on $\{e_t\}$ rejects the unit root with probability approaching 1 as T grows, regardless of whether y_t and x_t are cointegrated.

We do not prove this. The reasoning is by analogy: Eroğlu, Miller, and Yiğit [10] prove that QMLE time-varying-parameter state-space estimation absorbs the integrated error into the estimated state at the *cointegrating-vector-estimator* level. The screen we audit applies ADF one level downstream, to the one-step innovation sequence of the same filter under frozen parameters; transferring their absorption result to that innovation sequence is the natural analogue, and our placebos (Table 1, Table 2) and positive control (Table 3) are consistent with it. A formal transfer through the frozen forward-rolled filter's steady-state Kalman gain, with Said-Dickey regularity verified on the resulting innovations, is left for future work.

Diagnostic note on q_β. The fitted process variance is small (median $q_\beta \approx 2\times10^{-9}$ on the positive control). Even at very small $q_\beta > 0$ the filter operates in the steady-state-whitening regime [13], so the absorption mechanism still applies, and the placebo pass rate is the same at every q_β we observed.

Eroğlu, Miller, and Yiğit [10] also propose a correction. Their remedy operates at the cointegrating-vector level: an estimator that recovers a (possibly time-varying) cointegrating coefficient and distinguishes among no cointegration, fixed cointegration, and time-varying cointegration. This does not directly repair the screen-level pipeline we audit, which is the binary decision "does this pair pass the ADF screen?" The correct screen-level response to the artifact is what we run here: a matched placebo. Engineering a one-shot pass/fail filter that inherits their estimator's discrimination is left for future work.

An earlier draft of this project defended the result on the grounds that the parameters were fit on training data only and never refit on test. That defense misses the mechanism. The whitening holds out-of-sample too, because the filter keeps adapting on the test window from its frozen dynamics.

A placebo settles it. We ran the identical screen on series that cannot be cointegrated: (a) independent simulated random walks, (b) each coin paired with a phase-randomized surrogate of another coin, and (c) real pairs with one leg block-shuffled. All three pass at the same nearly 100% rate as the real pairs (Table 1). A clean, noncircular test tells the opposite story. Engle-Granger [9] on the test window, and a static OLS ADF, put genuine out-of-sample cointegration at 2 to 3% of pairs on the matched window, at or below the 5% null floor. At the critical values used in this pipeline, the screen does not separate real pairs from matched placebos. We are not claiming that no pair ever comoves, only that this screen, as run, cannot detect it.

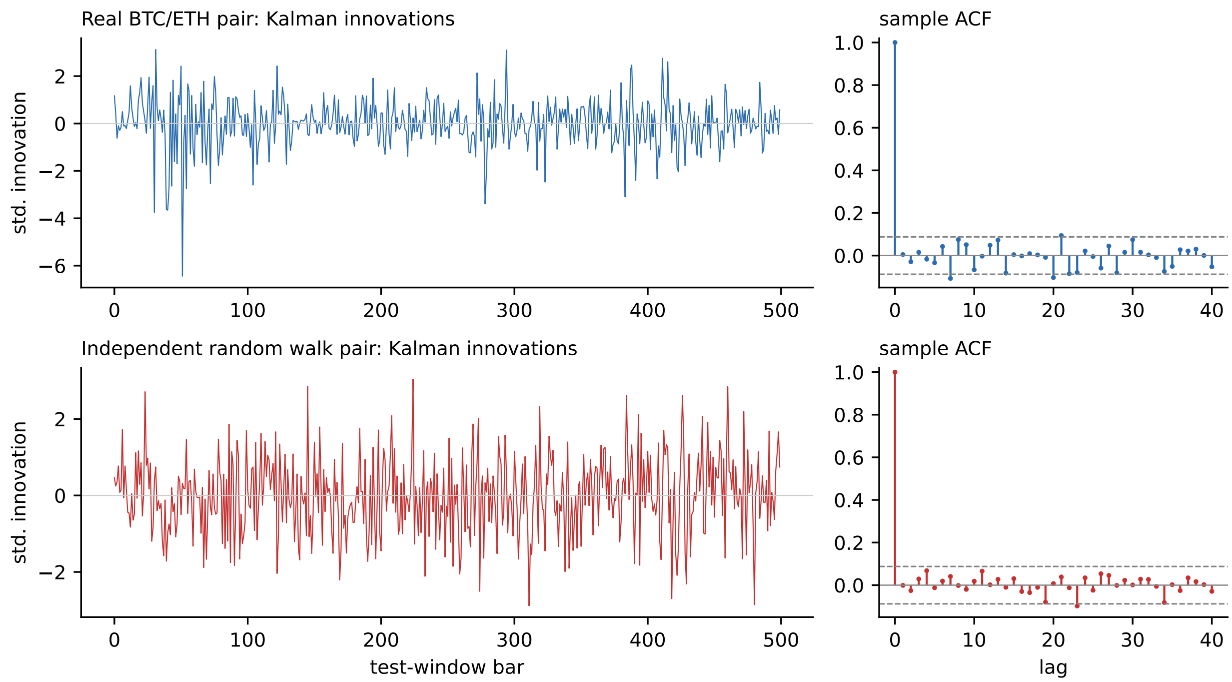

Figure 1: A Kalman filter whitens its own innovations: a representative real pair (top) and an independent random-walk pair (bottom) produce visually indistinguishable innovation series, with sample autocorrelation functions flat past lag 1. This is the mechanism behind the artifact documented in Table 1.

The artifact does not depend on that single start date. We re-ran the identical screen at five disjoint quarterly anchors from mid-2023 to mid-2024 (Table 2). At every one, real pairs and all three placebos pass the Kalman innovation ADF at 99 to 100%, with Wilson 95% intervals that never fall below 95%, while the clean Engle-Granger benchmark stays between 1 and 8%. The screen reports the same near-total cointegration whether the input can be cointegrated or not, at every start date we tried.

We also ran a positive control: sixty constructed pairs that are genuinely cointegrated, $y_t = x_t + e_t$ with x_t a random-walk and e_t a tight stationary AR(1). The Kalman screen passes them at 100%, the same rate it passes the random-walk negatives, so it does not separate true cointegration from none (Table 3). We can see the mechanism directly: the test window innovations are white, with a Ljung-Box test failing to reject whiteness for 87% of the cointegrated pairs and 95% of the random-walk pairs, while the static OLS residual stays autocorrelated for both. The fitted hedge barely moves (median process variance $q_\beta \approx 2 \times 10^{-9}$), so the whitening comes from the filter fitting noise rather than tracking a real relationship. The clean tests do separate the two, passing the positive control more often than the negative (17% versus 2% for Engle-Granger, 27% versus 8% for static OLS), though genuine out-of-sample cointegration is hard to detect on a short window.

The artifact does not depend on the bar. We reran the screen at hourly, four-hour, and daily frequencies on a recent multiyear window (Table 4). Real pairs and random-walk placebos pass together at every cadence (Figure 3): 100% versus 100% versus 100% at hourly and four-hour, and 70% versus 68% at daily ($p < 0.001$). The clean tests behave differently. Engle-Granger runs at 25% on this long hourly window and falls to the 5% null floor at daily. A static OLS ADF runs

Table 1: The Kalman screen passes placebos that cannot be cointegrated at the same rate as real pairs. Out of sample ADF on the Kalman innovations, top 50 USDT universe (250 pair subsample), $t_0 = 2024\text{-}01\text{-}01$, 90 day train / 30 day test. The 2 to 3% clean-test rate is at or below the nominal 5% test size, consistent with the null of no cointegration on this matched window.

Series tested	ADF $p < 0.05$	ADF $p < 0.001$
Real top 50 pairs	100.0%	96.4%
(a) Independent random-walks	100.0%	98.3%
(b) Coin vs. phase-randomized surrogate	100.0%	95.0%
(c) Real pair, one leg block-shuffled	100.0%	95.0%
Clean tests (EG and static OLS, test window)	2 to 3%	n/a

Table 2: The artifact is not specific to one start date: the identical screen at five disjoint quarterly anchors. Real pairs and all three placebos (independent random walk, phase-randomized, block-shuffled) pass the Kalman innovation ADF at $p < 0.05$ at nearly 100% (Wilson 95% intervals in brackets; $n \approx 120$ real, 80 per placebo), while the clean Engle-Granger test stays low. 90-day train / 30-day test at each t_0.

Anchor t_0	Real Kalman	All placebos	Clean EG
2023-07-01	100% $[97, 100]$	100%	3.8%
2023-10-01	99.2% $[95, 100]$	100%	1.4%
2024-01-01	100% $[97, 100]$	100%	2.1%
2024-04-01	100% $[97, 100]$	100%	8.1%
2024-07-01	100% $[97, 100]$	100%	6.4%

higher still, from 42% at hourly to 8% at daily. Both are window dependent. Over long hourly spans the crypto majors share a common trend, which a clean test reads as broad comovement, and which the matched short window of Table 1 puts back at the 2 to 3% floor. Neither clean test approaches the screen's 100%, and the small difference between the two tables' real $p < 0.001$ rates (96% versus 100%) is window and sample size, not a contradiction. At the critical values used in this pipeline, the Kalman screen does not separate real pairs from matched placebos at any frequency we tested.

4.2 The rolling z score

The second screen is a rolling z score, the standard entry signal for pair trades. For a spread s_t it subtracts a trailing mean and divides by a trailing standard deviation,

$$z_t = \frac{s_t - \bar{s}_{t-1}}{\sigma_{t-1}},$$

and it trades when $|z_t|$ is large, betting the spread returns to its mean. On our spreads it produced a large reversion edge.

The edge is mechanical. Subtracting a trailing mean from any series, including a pure random walk, manufactures mean reversion, because the trailing mean chases the level and the deviation from it cannot wander as freely as the raw series. Demeaning by a trailing window of length L induces negative autocorrelation of order $1/L$ in the result. This is the time-aggregation bias

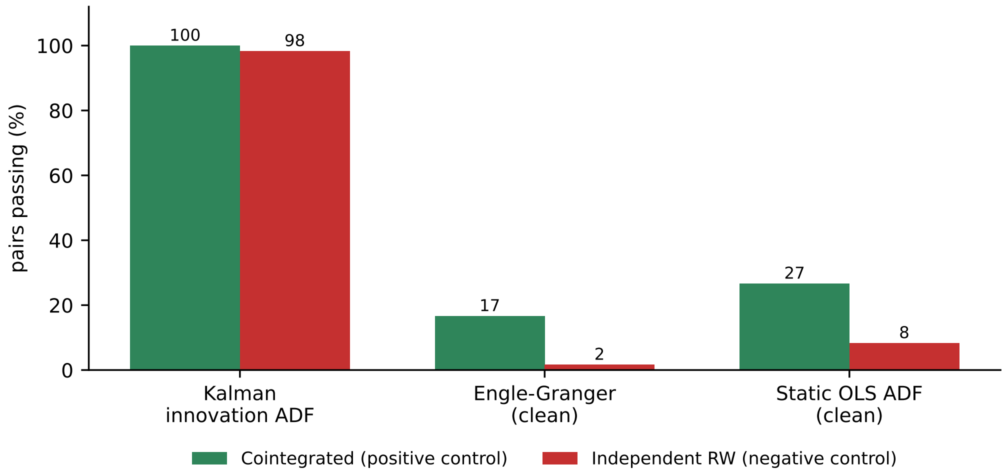

Figure 2: Positive vs negative control: the Kalman screen passes both the truly cointegrated pairs and independent random-walks at near 100%, while the clean Engle-Granger and static OLS tests separate the two groups. Visual companion to Table 3.

Table 3: Positive versus negative control. The Kalman screen passes both at the same rate and its innovations are white in both, so it does not separate them. The clean tests do. 60 constructed pairs per group, 1500 bar train, 500 bar test.

	Cointegrated	Random walks
Kalman innovation ADF passes ($p < 0.05$)	100.0%	98.3%
Kalman innovations white (Ljung-Box $p > 0.05$)	86.7%	95.0%
Static OLS residual white	0.0%	0.0%
Clean Engle-Granger passes	16.7%	1.7%
Clean static OLS ADF passes	26.7%	8.3%

documented by Working [30]: first differences of moving averages of a random chain are spuriously correlated even when the underlying chain is not. Cochrane [5]'s variance-ratio analysis of GNP gives the same lesson from the persistence side. The floor is therefore calculable rather than fitted. A random-walk has no mean to revert to, yet a rolling z score on one still reverts.

We measured the mechanical floor two ways, and both are large. Per $|z| > 2$ event, the signed reversion on pure random-walks is positive at every frequency we tested: $+0.98z$ at hourly, $+1.61z$ at four-hour, and $+1.51z$ at daily, each many standard deviations from zero (Table 5; Figure 4 shows the mechanism on one path). As a trading rule at hourly frequency, a rolling z strategy that posts a Sharpe near 4.75 on our data still posts about 2.2 on random-pairs and 2.4 on phase-randomized noise. Most of the apparent edge is the floor. For this reason every per-event reversion magnitude elsewhere in this paper is reported net of a per-pair, per window random-walk floor.

Both screens share a failure mode. They manufacture the appearance of structure that a matched placebo reproduces. Both are robust to the analysis choices: the rolling z floor stays positive ($+0.45$ to $+2.6z$) across lookback, horizon, and threshold, and the Kalman placebo passes at 85% or more across the ADF lag, so neither is an artifact of the parameters we picked. A cointegration screen built on a filter's own innovations must be checked against a white noise placebo before it is believed, and a rolling z reversion claim must be checked against a random-

Table 4: Frequency robustness of the Kalman artifact. Real pairs and random-walk placebos pass the screen together at every cadence. The clean Engle-Granger and static OLS benchmarks do not. Top-50 universe, 60 pairs per cell, recent multiyear window.

	Real pairs		RW placebo		Clean tests $p < 0.05$	
Frequency	$p < 0.05$	$p < 0.001$	$p < 0.05$	$p < 0.001$	Engle-Granger	static OLS
Hourly	100.0%	100.0%	100.0%	100.0%	25.0%	41.7%
4 hour	100.0%	100.0%	100.0%	100.0%	11.7%	20.0%
Daily	100.0%	70.0%	100.0%	68.3%	5.0%	8.3%

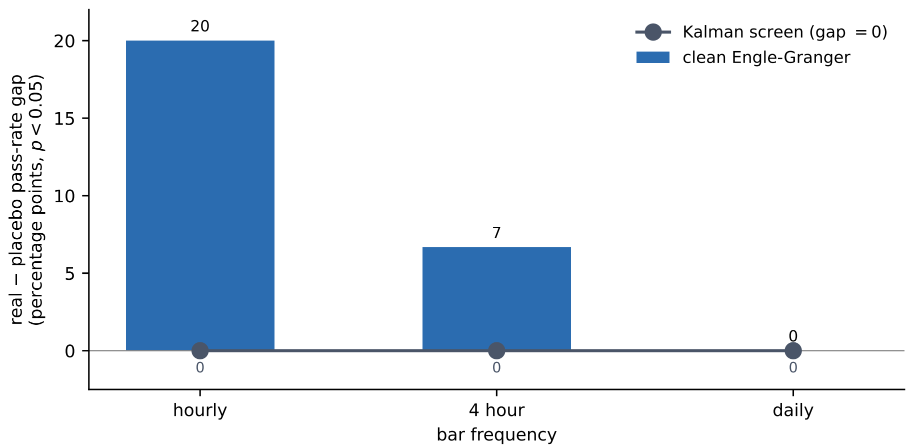

Figure 3: Placebo gap by frequency. The Kalman screen's (real minus random-walk placebo) pass rate gap is zero to within sampling noise at every cadence, while the clean Engle-Granger test's gap is large and frequency dependent. The clean test's random-walk placebo passes at its nominal 5% size. This reshaped view makes visible what Table 4's percentages only imply: a flat zero line is the diagnostic.

walk null. Neither check is standard practice, and without them both screens report a signal that is not there. This is the lesson the backtest-overfitting literature draws for strategy selection [3], applied one level earlier, to the screens that build the spread.

5 Real but untradable structure

Strip the two artifacts and some genuine structure remains. None of it is tradable. Either the market prices it faster than anyone could act on it, or it is a statistical regularity that does not turn into money. We take the clearest case of each.

5.1 Reversion-speed persistence

Mean reversion persists out-of-sample even though cointegration does not. Across nine disjoint walk forward splits (six month train, three-month test), a pair's in-sample OU reversion speed (Appendix A.6) predicts its out-of-sample *excess* reversion, net of the rolling z floor of Section 4.2: Spearman rank correlation $\rho = 0.46$ (95% CI $[0.37, 0.54]$, $p < 0.001$), positive in every one of the nine splits. The top quintile shows about $0.69z$ more excess reversion than the bottom out-of-

Table 5: The rolling z score manufactures mean reversion on pure random walks, at every frequency. Mean signed reversion per $|z| > 2$ event, 200 random-walk surrogates per cell.

Frequency	Mean event reversion (z)	s.d.
Hourly	+0.98	0.26
4 hour	+1.61	0.39
Daily	+1.51	0.45

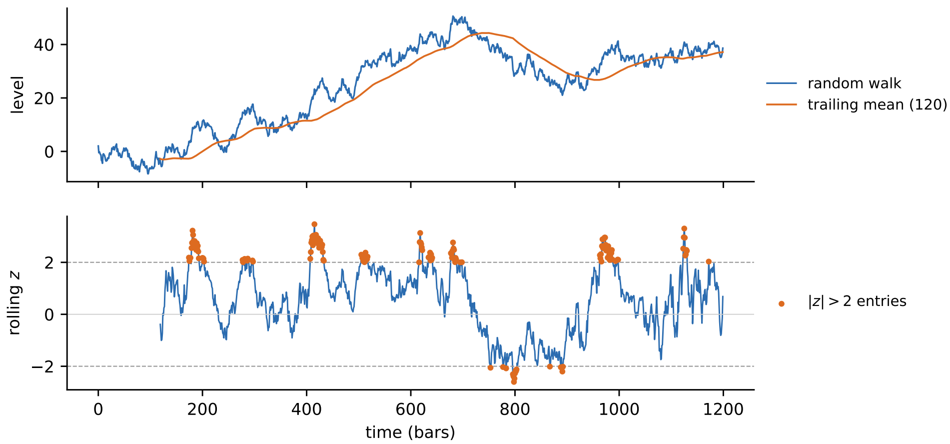

Figure 4: A rolling z score manufactures mean reversion on a pure random walk. Top: one simulated random-walk and its trailing mean. Bottom: the resulting rolling z, which crosses ± 2 and reverts toward zero, although the series has no mean to revert to.

sample ($[0.58, 0.81]$, $p < 0.001$). The placebos are clean: a random quintile split gives $+0.0001z$, and shuffling the train to test link collapses ρ to -0.002. Reversion speed is a real, graded, persistent property that ranks pairs.

One caveat belongs on that coefficient. The $\rho = 0.46$ is the strongest of four in-sample reversion metrics we tried, chosen after the fact, and it is not corrected for that choice. The direction holds across all nine splits. The exact coefficient is a best case.

Ranking pairs by reversion speed tells you where reversion lives, but not whether you can make money on it. Extracting the edge after costs is a separate question, and the strategy built on the property (Section 6) is marginal at best.

5.2 Order flow at tradable horizons

We tested the order flow ideas on a 2024 sample of Binance L2 book and trade data: aggregate book metrics on the top 50 USDT majors, and event level tests on four majors (BTC, ETH, SOL, AVAX) across days drawn from all twelve months, from Tardis. The book carries real, contemporaneous information about price formation, but none of it survives to a horizon you could trade against. That null holds for both forecasting and execution.

Trade size does proxy information. Large trades carry about twice the one second price impact of small ones, and the ranking replicates across all of 2024 on four coins (one second impact for BTC of 0.33/0.23/0.13 bps for institutional, mid, and retail, with institutional 2.0 to 2.6× retail, though the longer horizon permanent estimate is noisier; Table 6). Book side order flow imbalance [6]

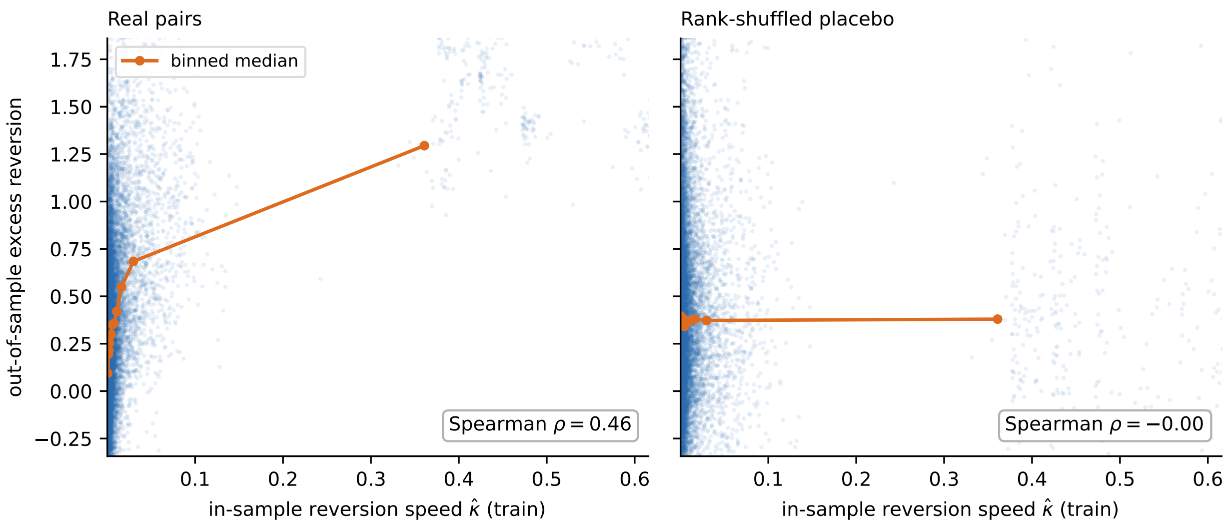

Figure 5: In sample OU reversion speed predicts out-of-sample excess reversion across 9 walk forward splits (Spearman $\rho = 0.46$). Left panel: real pairs; the positive slope is present in each of the nine splits. Right panel: rank shuffled placebo collapses to noise.

explains sub-second midprice moves better than trade flow does (contemporaneous incremental R^2 of +0.11 to +0.15 for BTC and ETH), and 77 to 90% of the liquidity that leaves the best quote is cancellations, invisible to trade only analysis (Figure 7).

Table 6: Per trade price impact at 1-second horizon by trade size bucket and symbol: institutional (>$10k) consistently 2.0 to 2.6× retail across BTC/ETH/SOL/AVAX, validating size as a per-order information proxy.

Symbol	Retail (bps)	Mid (bps)	Institutional (bps)
BTCUSDT	0.13	0.23	0.33
ETHUSDT	0.15	0.24	0.32
SOLUSDT	0.15	0.26	0.32
AVAXUSDT	0.18	0.38	0.38

All of it decays within seconds. The predictive content of order flow falls to near zero by 30 seconds (Figure 6), and at a 15-minute to one hour horizon institutional flow predicts a mild reversal, not continuation, with a negligible incremental R^2. Hidden and iceberg liquidity is measurable (about 1 to 4% of at best volume, with iceberg episode rates higher) but carries no edge.

We then measured whether the signal helps execution, the usual fallback when a signal does not forecast returns. It does not (Table 7). Across 29,952 simulated parent orders over event level book and tape, aggressive crossing is the cheapest method for these majors, a signal-timed passive rule does worse than a random one at the same posting rate, and the signal's correlation with the per-order advantage of posting over crossing is +0.03, noise. A perfect foresight oracle would save 1.4 bps, so the opportunity is real, but no contemporaneous book feature forecasts it. The null holds across all twelve months of 2024, including the August 5 crash. Schnaubelt [24] reaches the opposite headline with a different metric and a different agent class: an RL placement policy trained on 3.5M order book states across 18 months posts a 37.71% implementation-shortfall reduction vs immediate

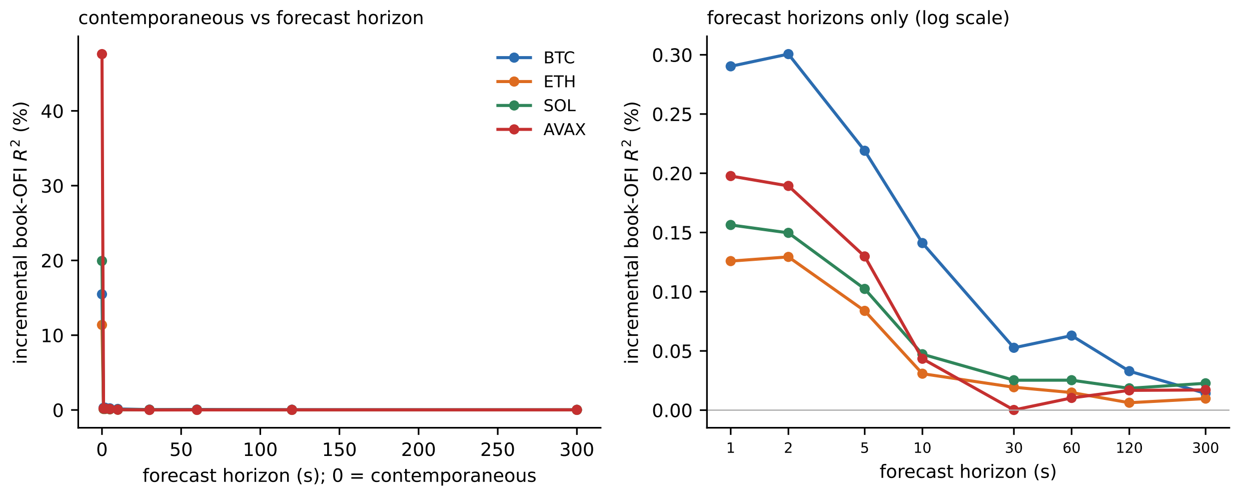

Figure 6: Predictive incremental R^2 of book-OFI for forward midprice moves decays from ∼11 to 48% contemporaneously down to under 0.3% at a one second forecast horizon and a 0.01 to 0.06% noise floor out to 300 seconds, across BTC/ETH/SOL/AVAX and all 2024 market regimes. The information half-life is seconds; no tradable forecast horizon survives.

market execution. The two findings are not in conflict. His agent learns a state-conditional mixture of posting and crossing on a different basket and a different cost decomposition; ours measures whether any single-shot contemporaneous L2 feature realizes the oracle-bounded passive-vs-cross advantage on our basket. The complementarity is the oracle bound: at 1.4 bps it caps how much any contemporaneous-feature rule on majors can save against always-crossing, and a learned policy must work mainly through state conditioning, not the static L2 signal. Table 8 confirms the pooled result holds per symbol and per order size.

Table 7: Order flow signals do not improve execution. Implementation shortfall versus arrival mid, pooled, 30 second horizon. Lower is cheaper. 29,952 orders, BTC/ETH/SOL/AVAX.

Execution rule	Cost vs. arrival mid (bps)
Aggressive (always cross)	+1.10
Naive passive (always post)	+2.69
Signal timed (post/cross on the L2 signal)	+1.73
Random at the same posting rate	+1.68
Oracle (perfect foresight)	−0.30

The one useful byproduct is a cost number. Walking the real book puts execution costs about 17 to 23% below the flat 5 bps slippage component the earlier backtests assumed. Cheaper costs do not rescue the strategy.

The efficiency reading is scoped to the liquid core. The L2 tests cover top 50 aggregate book metrics and four megacap symbols on the global venue. The thin Binance.US microcaps where the marginal reversion edge of Section 6 lives are not in this sample. The two findings plausibly fit together, though across nonoverlapping samples rather than on one venue: the majors we can measure are efficient, and the names with an apparent edge are too thin to trade. We did not measure the microcaps' microstructure directly, so the join is an inference.

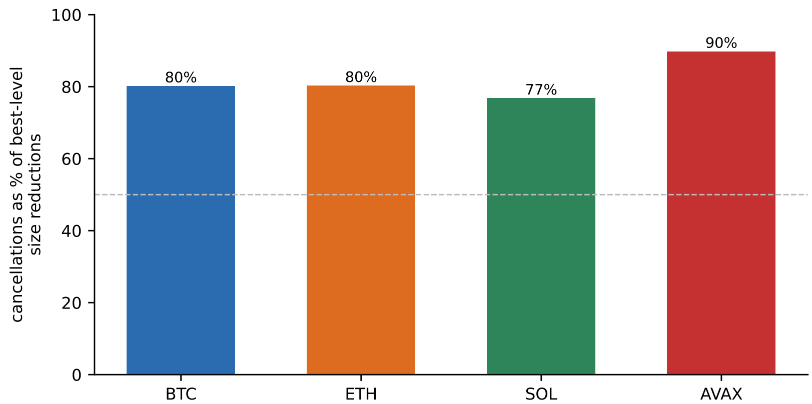

Figure 7: Cancellations dominate trades in best level liquidity withdrawal: 77 to 90% (about 77 to 80% for BTC, ETH, SOL; 90% for AVAX) of best level size reductions are cancels, invisible to trade only microstructure analysis (BTC/ETH/SOL/AVAX, 2024).

Table 8: Per symbol breakdown of execution costs at $10k and $50k notional. Aggressive crossing dominates across all four majors and both sizes: the L2 signal-timed rule is more expensive than crossing in every cell, consistent with the pooled result of Table 7.

	$10k notional (bps)		$50k notional (bps)	
Symbol	Aggressive	Signal timed	Aggressive	Signal timed
BTCUSDT	0.02	0.64	0.09	0.71
ETHUSDT	0.08	0.80	0.26	0.94
SOLUSDT	0.56	1.36	1.28	2.01
AVAXUSDT	2.09	2.70	4.45	4.69

6 A marginal survivor

One strategy survives, but only barely. There is a real, market-neutral, diversified mean-reversion effect in crypto pair spreads. Its tradable magnitude is uncertain, it exists only without a stop, and a literal backtest of it inflates spectacularly. We start with the limits, since they matter more than the headline number.

6.1 Construction and market-neutrality

Selecting pairs by reversion speed and holding the spread to convergence, with a static train window hedge and z score (not the rolling z of Section 4.2), produces a positive, market-neutral return. It is diversified: beta to an equal-weight crypto index is about -0.06, and the top ten principal component portfolios explain only 1.7% of its return, so it is not a hidden factor bet. It replicates on an independent exchange (Coinbase, USD quoted) and under four from-scratch reimplementations that share none of our pipeline, none of which found look ahead, so the effect is not a coding artifact.

6.2 Sensitivity to the exit rule

The exit rule decides the sign of the P&L. With a conventional $|z| = 4$ stop the strategy loses (Sharpe -2.25), and the stop is the cause: it realizes losses on spreads that then revert. Remove the stop and hold to convergence and the strategy turns positive. The project set out to build an hourly strategy with tight risk management. This is the opposite: a patient book that never closes on a stop, with a median hold near 35 days, in which 78% of positions do not converge within a quarter and about a third of the return is a generic survivor comovement floor.

6.3 Magnitude and the preregistered held-out test

The naive Binance Sharpe was first reported near 2.5, and four independent reimplementations put it at 2.0 to 2.2. Neither is the honest figure. Month long holds make hourly returns heavily autocorrelated (Figure 8), so a Newey-West correction [22] drops the monthly Binance number to about 1.4, and on the independent Coinbase venue it is about 0.85 to 0.90. The HAC- and venue-honest figure is near 1.0 (Table 9), and the edge concentrates in recent, high-volatility periods.

Table 9: Naive vs HAC adjusted monthly Sharpe on Binance.US and Coinbase for the no-stop reversion strategy. Reading: the venue gap ($\sim$2.2 Binance vs $\sim$0.88 Coinbase) and the serial correlation adjustment (Binance naive 2.18 $\rightarrow$ HAC 1.4) together center the honest figure at $\sim$1.0. The Coinbase HAC entry is not separately computed; the naive Coinbase figure is shown unchanged. The $\sim$1.0 centered figure is a venue-midpoint heuristic, not a properly HAC-adjusted cross-venue estimate.

Series	Naive monthly Sharpe	HAC adjusted (monthly)
Binance.US, hourly returns	2.18	≈ 1.4
Coinbase, hourly returns	0.88	0.88
HAC- and venue centered	n/a	≈ 1.0

A preregistered, held-out test makes the fragility concrete. We locked the configuration before running (Section 3 records the lock) and evaluated it once on held-out data, from June 2025 to May 2026. The literal result was a monthly Sharpe near 5, *higher* than the in-sample figure, and it is not a tradable edge. It is inflation spread broadly across high-volatility microcap alts: individual pair windows post +300 to +517% in three-months under a flat 15 bps cost that badly understates the true cost of trading those names, with unit notional accounting that ignores capacity. A +517% three-month leg implies roughly a 150$\times$ gross return, not realizable at any size. Those names are barely traded (Figure 9): on Binance.US the held-out microcaps (WAXP, ZIL, VTHO, ACH, EGLD) turn over a median of tens to a few thousand dollars a day, roughly three to four thousand times less than the majors, so the deployable notional is negligible and a flat, size-independent cost is indefensible. The same machinery on a venue without those microcaps returns the ~ 1.0 figure. So the held-out test does not confirm an alpha. It shows the raw backtest is untrustworthy, and our own surviving result fails the same test the rest of the paper applies to everything else.

The preregistration had a second arm, which we report in full. A single selection over all data before June 2025 qualified only two pairs (long-horizon reversion in the tradable band is rare), and they posted a monthly Sharpe of 1.60. The HAC-corrected, hourly-annualized Sharpe on the same series is 0.95 with 95% confidence interval $[-0.21, 1.89]$, which spans zero. The unit change (monthly vs hourly-annualized) reflects the inference: the monthly statistic does not have enough observations on a single 11-month tail to compute a defensible HAC interval, so the confidence

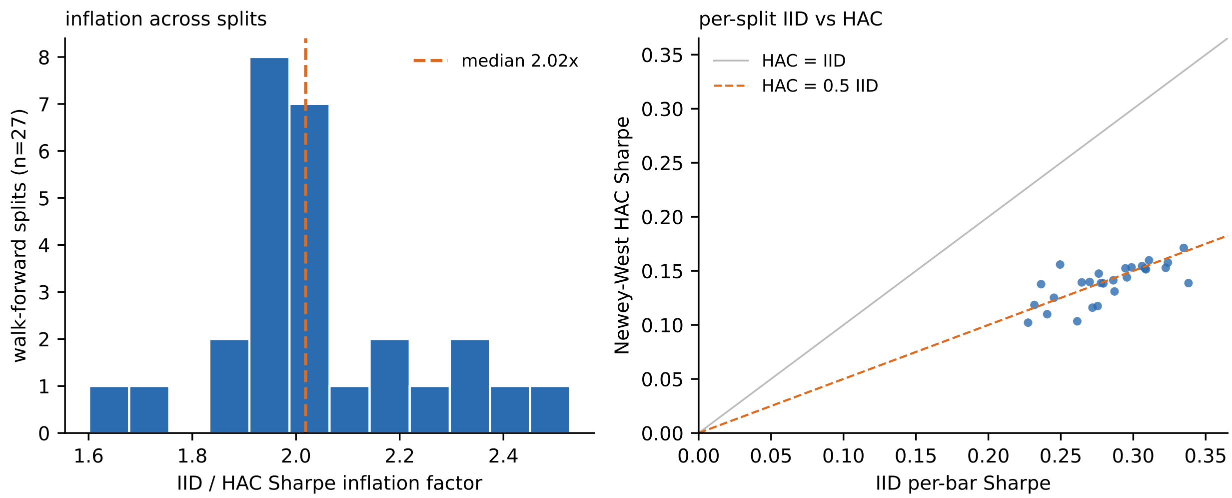

Figure 8: Hourly Sharpe is heavily serial correlation inflated by multiweek holds. Left: per split IID vs Newey-West HAC ratios cluster around 2× (median 2.0) across the 27 walk forward splits. Right: scatter of IID vs HAC Sharpe per split, with the HAC $= 0.5 \cdot$ IID reference line.

statement is reported on the hourly-annualized series where HAC at lag 240 is well-defined. Either way, the test confirms the inflation and gives no significant evidence for a deployable edge. The ~ 1.0 figure is itself exploratory, adjusted for HAC and venue after the search, and was never preregistered.

6.4 Conditions for deployability

The effect is also selection sensitive and survivorship fragile. A deflated Sharpe correction (Section 3) survives the no-stop family of configurations but fails the whole stop versus no-stop search, and the supports we might cite for it (the t statistic, the $\rho = 0.46$ of Section 5, the market-neutrality) share the same universe, selection, and windows, so they are correlated rather than independent confirmations. Its survivorship robustness rests on the selector avoiding coins that later delist, not on surviving them: a held position in a coin that collapses is a -100% leg. We tested this directly by forcing the book to hold through three real collapses (LUNA, UST, FTT, each paired with BTC, with the selector's avoidance overridden, Figure 10). Without a stop the loss runs -117% to -258% per-pair, with intracollapse drawdowns to $-1{,}300\%$. The structural-break circuit breaker (halt after a single leg moves more than 50% against the position) caps the loss to about -65% to -91% and the worst drawdown to -117%. The 50% threshold is the tightest level at which monthly Sharpe stays within 2% of the no-breaker baseline: across a sweep $K \in \{25, 30, 40, 50, 60, 70, 80, 90\}\%$, Sharpe retention drops to 0.85–0.94 for $K \leq 40\%$ (the breaker fires on noise and exits real reversion) and stays at $\sim$1.00–1.02 for $K \geq 50\%$, while the worst per-pair loss climbs from -131% at $K = 50\%$ to -207% at $K = 90\%$. So 50% is the sweet spot, not an arbitrary cutoff. The breaker blunts the tail but does not remove it, and at the book level it is what separates a bounded delisting risk from an unbounded one. We also inject permanent delisting scale breaks into a fraction of the 40 pairs each quarter (Figure 11). Without the breaker the no-stop book's monthly Sharpe falls to zero and then negative as the break rate rises. With the breaker it stays positive across the whole range, keeping about 96% of its monthly Sharpe at an 8%/yr break rate and 88% at 19%/yr.

A few things make those retention figures optimistic. First, the injected breaks ramp smoothly,

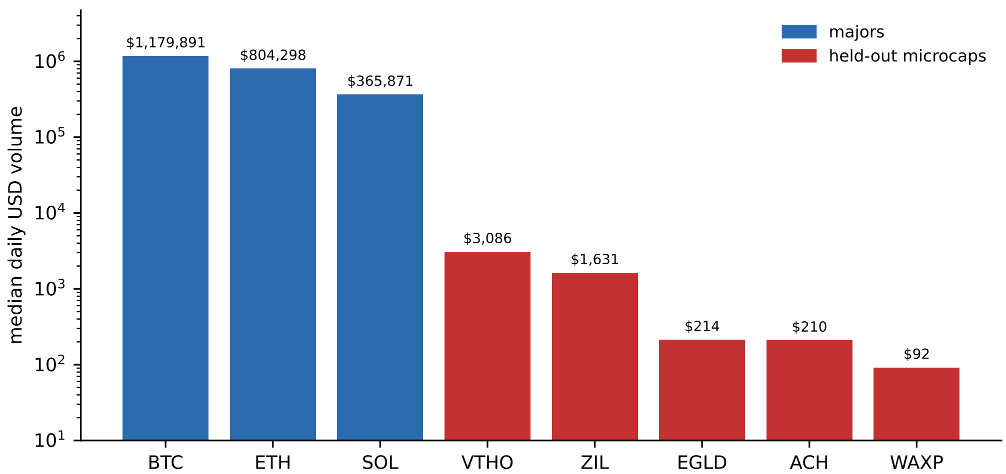

Figure 9: Median daily USD volume on Binance.US, log scale. The preregistered held-out microcaps (WAXP, ZIL, VTHO, ACH, EGLD; red) trade at tens to a few thousand dollars per day, three to four orders of magnitude below the majors (blue) whose costs the backtest assumes. A flat, size-independent transaction cost is therefore indefensible on these names.

so the breaker exits near its 50% threshold, whereas the real gapping collapses above cost it −65% to −91% per-pair. Second, the figures sit on the gross book, not the HAC- and venue honest ∼ 1.0, against which a delisting drag would bite proportionally harder. Third, the breaks are drawn independently, which understates the way breaks cluster in market wide selloffs. The robust finding is the qualitative one: the circuit breaker, not diversification alone, is what bounds the delisting tail, and at the low single digit yearly attrition that liquid top 50 names actually see, the haircut is modest. A fully point-in-time universe of every delisted coin is still beyond our data.

The defensible claim is therefore the sensitivity itself. There is a real, market-neutral reversion signal in crypto pair spreads that can be selected by reversion speed. Whether it is worth anything after realistic costs, capacity, and selection is unresolved, and the most honest single number, a monthly Sharpe near 1.0, is an estimate that a literal backtest does not cap from above.

7 Limitations

Five limitations bound these results, and the survivorship gap is the one most likely to change the conclusion.

- *Survivorship.* The local store holds currently listed symbols, so most delisted coins are absent. The surviving strategy's robustness rests on the selector avoiding coins that later delist. Forced to hold through three real collapses (Section 6) the no stop loss runs −117% to −258% per-pair and the circuit breaker caps it near −65% to −91%. A complete point-in-time universe of every delisted coin is still beyond our data.
- *Selection.* No single configuration was fixed before the search and evaluated once across the whole project. The preregistered test of Section 6 covers one configuration but cannot undo the selection that produced it, so the deflated Sharpe caution stands.
- *Microstructure coverage.* The event-level L2 tests use four symbols (BTC, ETH, SOL, AVAX)

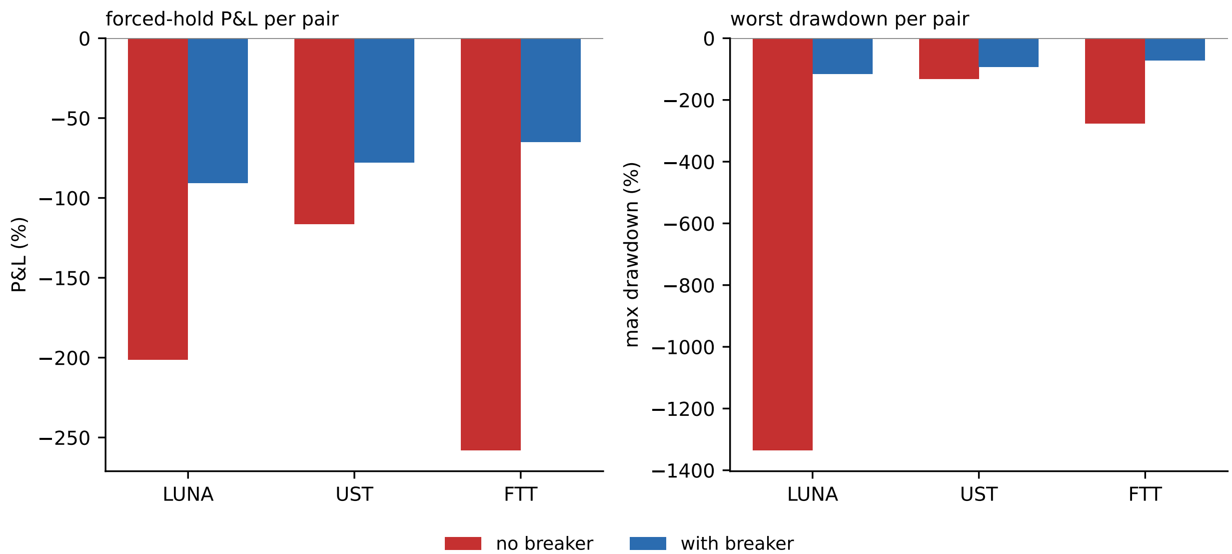

Figure 10: Per pair P&L when the no-stop rule is forced to hold through three real delisting collapses (LUNA-May 2022, UST-May 2022, FTT-Nov 2022). Without the circuit breaker, losses run −117% to −258% per-pair (a leg can lose more than 100% on the log spread accounting); with the 50% single leg breaker, losses are capped at −65% to −91%.

over days sampled across 2024, not a continuous year on the full universe. The aggregate book metrics use the top 50. The execution and impact results are four symbol.

- *Single venue and quote.* The spread results are Binance.US, USDT quoted, with a Coinbase USD-quoted cross check on the overlapping liquid majors. The microstructure is Binance.com (global) spot. The cross-check is not a same-instrument replication: USD-quoted Coinbase pairs have a different participant base, different fee schedule, and different microstructure from USDT-quoted Binance.US pairs, so part of the Coinbase-Binance Sharpe gap is venue and quote currency rather than overfitting. We have not tested other quote currencies or a broader set of venues.

- *Costs and capacity.* Costs are a flat per-leg charge. We do not model market impact or capacity, which is exactly what inflates the held-out microcap backtest of Section 6. The corollary is that every Sharpe in the paper is a unit-notional, capacity-unbounded figure: the no-stop reversion strategy's ∼1.0 HAC-and-venue-corrected Sharpe is the right order of magnitude for the qualitative claim but is not a defensible estimate of what a sized book would earn after realistic impact. An impact-aware sized run is the constructive next step we have not done.

- *Execution-simulator fill model.* The L2 execution test of Section 5 runs 29,952 simulated parent orders against the historical book, but the fill model is intentionally simple: a posted limit order is filled when the inside quote crosses it within the test horizon, with no queue-position tracking, no latency-to-cancel budget, and no separate adverse-selection-conditional-on-fill treatment. This understates the true cost of passive execution (good fills are taken first; sticky orders accumulate adverse selection), so the negative finding for signal-timed posting is robust to making the fill model more realistic. The perfect-foresight oracle is unaffected because it is computed under the same fill model.

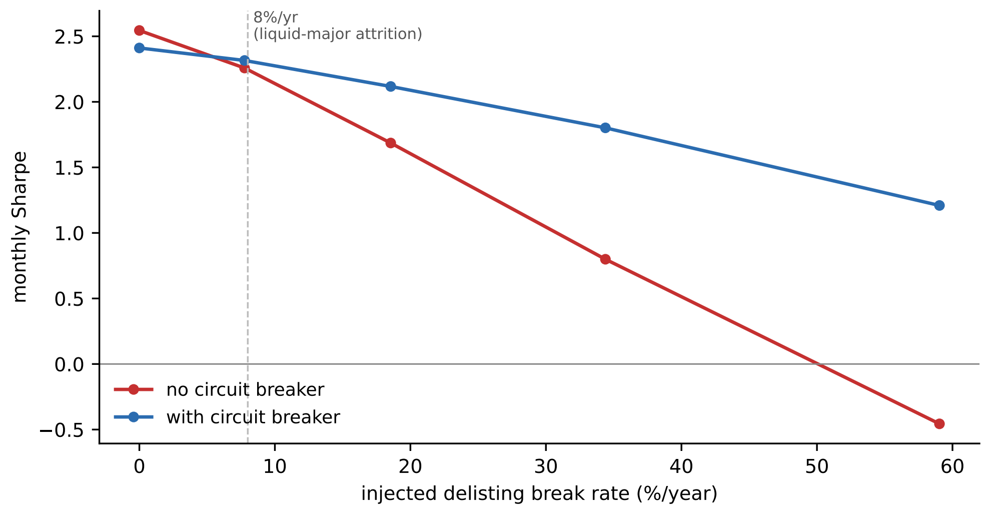

Figure 11: Monthly Sharpe of the no-stop reversion book as a function of injected delisting rate, with and without the structural-break circuit breaker. Without the breaker, the Sharpe declines steadily and crosses zero only near a 50%/year break rate. With the breaker, it retains ≈ 96% of its Sharpe at the 8%/year rate consistent with historical liquid major attrition.

8 Conclusion

These negative results say more about the screens we used than about the market itself. The Kalman cointegration test and the rolling z score both pass placebos they should fail, at every frequency we tested. Order flow forecasts only the next few seconds, a horizon no one can trade. The one market-neutral reversion effect we find is not statistically distinguishable from zero on the preregistered test (CI spans zero); even our exploratory point estimate near Sharpe 1.0 is marginal, stop sensitive, and selection sensitive, and a literal backtest of it inflates on costs it does not model.

The lesson is a methodological habit rather than a tradable strategy. Trust a cointegration or mean reversion screen only after you have shown it fails on a matched placebo built to reproduce the claim if the claim is mechanical. For crypto the evidence points to microstructure efficiency in the liquid majors at tradable horizons: the information is real, and the market has already priced it.

What would move the result is a genuinely point-in-time universe with delisted coins, a capacity and impact aware cost model, and a single preregistered configuration evaluated once on data the search never saw. Until then, the efficiency reading holds only for the liquid core we can measure. For the thin tail, where delisting risk concentrates, we cannot yet say.

Replication

Every quantitative claim in the paper is reproducible from the public repository at `https://github.com/samsam324/math-199-research`. The two screening-artifact placebos (Section 4) are in `scratch/audit_part1.py`, `audit_part1b.py`, and `audit_part2.py`; the positive-control table (Table 3) is in `scratch/kalman_positive_control.py`; the 50% circuit-breaker sensitivity sweep (Section 6) is in `scratch/threshold_sweep.py`; the survivorship analysis with delisted coins is in `scratch/t2_survivorship_rerun.py` and `t2_worstcase.py`; the L2 execution test (Table 7) is in `scratch/exec_value_2024.py`. The figures in this paper were produced by `scratch/make_figures_v2.py`.

The L2 dataset (Tardis Binance.com 2024) is in S3; query helpers are in `scripts/l2_query.py` and a presigned-link generator is in `scripts/presign.py`. All stochastic procedures (the 200 random-walk surrogates, 60 constructed positive-control pairs, phase-randomized and block-shuffled placebos, the 29,952 simulated parent orders, block-bootstrap resamples) use fixed seeds set at the top of each script (default `seed=42` unless documented otherwise in the script header); rerunning a script produces the same number to the rounding precision reported in the paper.

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A Methods background

This appendix collects working definitions of the tools used in the body of the paper. It is meant for a reader who has seen probability and basic time series but may not have used all of these specific instruments. We keep proofs out and give the formulas that are actually invoked in the text. Section references in parentheses point to where the tool is used.

A.1 Kalman filter and state-space form for dynamic hedge ratios

Let y_t be the dependent log price and x_t the independent log price in a pair. The dynamic hedge model treats the intercept α_t and slope β_t as latent states that drift over time. The state-space form is

$$\begin{aligned} \theta_t &= \theta_{t-1} + w_t, \qquad w_t \sim \mathcal{N}(0, Q), \\ y_t &= H_t\,\theta_t + v_t, \qquad v_t \sim \mathcal{N}(0, R), \end{aligned}$$

with $\theta_t = (\alpha_t, \beta_t)^\top$, observation matrix $H_t = (1, x_t)$, state-noise covariance $Q \in \mathbb{R}^{2\times 2}$, and observation-noise variance $R > 0$. The states follow a random walk; only y_t is observed.

Given the prior state mean $\hat{\theta}_{t-1|t-1}$ and covariance $P_{t-1|t-1}$, the Kalman recursion predicts

$$\hat{\theta}_{t|t-1} = \hat{\theta}_{t-1|t-1}, \qquad P_{t|t-1} = P_{t-1|t-1} + Q,$$

forms the one-step innovation

$$e_t = y_t - H_t\hat{\theta}_{t|t-1}, \qquad S_t = H_t P_{t|t-1} H_t^\top + R,$$

computes the Kalman gain

$$K_t = P_{t|t-1} H_t^\top S_t^{-1},$$

and updates

$$\hat{\theta}_{t|t} = \hat{\theta}_{t|t-1} + K_t e_t, \qquad P_{t|t} = (I - K_t H_t) P_{t|t-1}.$$

The gain K_t is the optimal blend of model prediction and new observation. The hyperparameters Q and R are estimated by maximizing the Gaussian innovation likelihood

$$\log \mathcal{L}(Q, R) = -\tfrac{1}{2} \sum_t \left(\log 2\pi S_t + e_t^2 / S_t\right),$$

on a training window, then frozen and forward-rolled on the test window so no test-window information leaks into the filter.

The mechanism behind the artifact in Section 4.1 lives in the innovation sequence. By construction, $\{e_t/\sqrt{S_t}\}$ has zero mean, unit variance, and zero serial correlation under the model, and at steady state the filter approaches this whitening behavior for any input [13]. Running a stationarity test on e_t therefore measures how well the filter has been optimized, not whether y_t and x_t share a common stochastic trend. A test that rejects the unit root on e_t is consistent with no cointegration whatsoever.

A.2 Augmented Dickey-Fuller test

The augmented Dickey-Fuller (ADF) test of Said and Dickey [23] tests the null H_0: the series z_t has a unit root (is $I(1)$) against H_1: z_t is stationary. The test regression is

$$\Delta z_t = \mu + \rho\, z_{t-1} + \sum_{i=1}^{p} \gamma_i\, \Delta z_{t-i} + \varepsilon_t,$$

with the augmentation lags included to soak up short-run serial correlation in ε_t. The test statistic is the t-ratio on $\hat{\rho}$, compared against Dickey-Fuller critical values. Under H_0, $\rho = 0$ and the process is a random walk; under H_1, $\rho < 0$ and shocks decay geometrically. Reject for sufficiently negative $t_{\hat{\rho}}$.

The connection to the Kalman innovation artifact (Section 4.1) is mechanical: a series that is close to white noise has trivially fast mean reversion, so the ADF rejects easily even when no economic mean-reverting relation is present. The test answers "is this stationary", not "is this a cointegrating residual".

A.3 Engle-Granger cointegration

Two integrated series $y_t, x_t \sim I(1)$ are cointegrated if there exists a constant β such that $y_t - \beta x_t$ is $I(0)$. The Engle-Granger two-step procedure [9] fits this β by static OLS on levels,

$$y_t \;=\; \alpha + \beta x_t + u_t,$$

forms the residuals $\hat{u}_t = y_t - \hat{\alpha} - \hat{\beta} x_t$, and applies an ADF test to $\hat{u}_t$ with Engle-Granger critical values (which are more conservative than ADF because $\hat{\beta}$ has been estimated). Rejection is evidence that the linear combination defined by $(\hat{\alpha}, \hat{\beta})$ is a stationary error-correction term.

This is the clean comparison for the Kalman-innovations procedure of Section A.1. Engle-Granger tests a residual whose dynamics are inherited from the original prices; the Kalman procedure tests an innovation sequence whose dynamics are inherited from the filter. The latter rejects the unit root on placebos that, by construction, cannot be cointegrated; the former does not.

A.4 Newey-West HAC standard errors

Ordinary OLS standard errors assume the regression errors are independent and homoskedastic. In time series with overlapping returns, slow-moving regressors, or autocorrelated targets, both assumptions fail and naive standard errors understate sampling uncertainty, often by a large factor. The HAC covariance estimator of Newey and West [22] corrects this. For a regression with score g_t, the Newey-West variance estimator is

$$\hat{\Sigma}_{NW} \;=\; \hat{\Gamma}_0 + \sum_{\ell=1}^{L} w_\ell \left(\hat{\Gamma}_\ell + \hat{\Gamma}_\ell^\top\right), \qquad w_\ell \;=\; 1 - \frac{\ell}{L+1},$$

where $\hat{\Gamma}_\ell = T^{-1} \sum_{t=\ell+1}^{T} g_t g_{t-\ell}^\top$ is the sample autocovariance at lag ℓ and w_ℓ is the Bartlett kernel. The kernel is non-negative and linearly decreasing in ℓ, which guarantees a positive semi-definite estimate.

We use $L = 60$ for one-hour analyses with a 60-bar target, $L = 24$ for daily analyses with a one-day target, and $L = 240$ for monthly Sharpe inference where ten daily lags buffer the monthly aggregation. Sharpe ratios reported as "HAC adjusted" divide the mean by the square root of the HAC variance of the mean, then scale to monthly.

A.5 Deflated Sharpe Ratio

The deflated Sharpe ratio (DSR) of Bailey and López de Prado [3] discounts a reported Sharpe for non-normal returns and for the number of strategies that were tried before the one that was reported. Let $\widehat{SR}$ be the in-sample Sharpe of the selected strategy, T the sample length, γ the sample skewness, and κ the sample kurtosis. The DSR is

$$\mathrm{DSR} \;=\; \Phi\left(\frac{(\widehat{SR} - SR_0)\sqrt{T-1}}{\sqrt{1 - \gamma\,\widehat{SR} + \frac{\kappa-1}{4}\,\widehat{SR}^2}} \right),$$

where Φ is the standard normal CDF. The threshold SR_0 is the expected maximum Sharpe under the null of zero true edge across N independent trials,

$$SR_0 \;=\; \sqrt{V[\,\widehat{SR}_n\,]}\,\big((1-\gamma_E)\,\Phi^{-1}\big(1 - 1/N\big) + \gamma_E\,\Phi^{-1}\big(1 - 1/(Ne)\big)\big)\,,$$

with $\gamma_E \approx 0.5772$ the Euler-Mascheroni constant and $V[\widehat{SR}_n]$ the cross-trial variance of estimated Sharpes under the null.

N matters because the maximum of N noisy estimates grows with N: trying ten strategies and reporting the best one has a much higher null Sharpe than trying one. Doubling the number of trials raises SR_0 by roughly $\sqrt{\log N}$ in the relevant regime. We use N equal to the number of distinct parameter or pair configurations evaluated before the reported one.

A.6 Ornstein-Uhlenbeck process and reversion speed

The Ornstein-Uhlenbeck (OU) process is the canonical continuous-time mean-reverting model. The SDE is

$$dS_t \;=\; \kappa\,(\mu - S_t)\,dt + \sigma\,dW_t,$$

with long-run mean μ, reversion speed $\kappa > 0$, instantaneous volatility σ, and W_t a Wiener process. The conditional mean decays geometrically toward μ with rate κ, so the expected time for a deviation to halve is the half-life

$$t_{1/2} \;=\; \frac{\ln 2}{\kappa}.$$

We estimate κ from a discretized AR(1) fit on the spread, $S_{t+1} - S_t = a + b(S_t - \mu) + \eta_t$ with $\hat{\kappa} = -\ln(1+b)/\Delta t$, and rank pairs by this estimate.

The finding in Section 5 that train-window $\hat{\kappa}$ predicts test-window $\hat{\kappa}$ at Spearman $\rho = 0.46$ across walk-forward splits is a statement about the process, not about profitability.

A.7 Walk-forward cross-validation

Walk-forward cross-validation is the non-anticipative analogue of k-fold CV for time series. Fix a train length T_{tr}, a test length T_{te}, and a step length T_s. Split i uses observations $[t_i, t_i + T_{\text{tr}})$ for fitting and $[t_i + T_{\text{tr}}, t_i + T_{\text{tr}} + T_{\text{te}})$ for evaluation, then advances $t_{i+1} = t_i + T_s$. Any parameter is refit per split using only the train window. Results are reported as the distribution across splits, not as a single in-sample number.

We use $T_{\text{tr}} = 6$ months, $T_{\text{te}} = 3$ months, and $T_s = 3$ months in most of the paper, with 90/30 in earlier microstructure work. Random k-fold is inappropriate for time series because it routinely places train observations after test observations and destroys the no-lookahead guarantee.

A.8 Placebo construction

A placebo is a synthetic series with a known absence of the structure being tested. Different nulls preserve different features, and the right placebo is the one that matches everything in the data except the mechanism in question. We use four.

Random walks via cumulative sums. Sample $\eta_t \sim \mathcal{N}(0, \sigma^2)$ i.i.d. and set $z_t = z_{t-1} + \eta_t$. This preserves nothing about the original series except the innovation variance; two independent random-walks share no common trend.

Phase randomization. Take the discrete Fourier transform $\tilde{z}_k = \mathcal{F}[z_t]$, replace each non-DC phase by a uniform draw on $[0, 2\pi)$ subject to Hermitian symmetry, and invert. The resulting series has the same power spectrum as the original but random phase, so all linear second-order statistics are preserved while any non-linear or cross-series alignment is destroyed.

Block shuffle. Partition z_t into contiguous blocks of length L and concatenate the blocks in a random order. Short-range autocorrelation up to lag $\sim L$ is preserved; long-range structure including unit roots and cointegrating drift is destroyed.

Random-pair pairing. Keep each price series intact but pair it with the price series of an unrelated coin. Marginal distributions, autocorrelations, volatility clustering, and trends are all preserved leg by leg; only the joint cointegration is destroyed.

The four nulls answer different questions. The Kalman screen of Section 4.1 fails the random-walk, phase-randomized, and block-shuffled nulls at the same rate as the real data, which is the diagnostic.

A.9 Microstructure terms

Mid quote. $p_t^{\text{mid}} = (p_t^a + p_t^b)/2$, with p_t^a the best ask and p_t^b the best bid.

Quoted spread. $s_t = p_t^a - p_t^b$. The relative spread is s_t/p_t^{mid}.

Microprice. The size-weighted mid of Stoikov [25],

$$p_t^{\text{micro}} \;=\; \frac{q_t^b\, p_t^a + q_t^a\, p_t^b}{q_t^a + q_t^b},$$

with q_t^a, q_t^b the displayed sizes at the top of book. The microprice tilts toward the side with less depth.

Order book imbalance. For top K levels,

$$\text{OBI}_t^{(K)} \;=\; \frac{\sum_{k=1}^K q_{t,k}^b - \sum_{k=1}^K q_{t,k}^a}{\sum_{k=1}^K q_{t,k}^b + \sum_{k=1}^K q_{t,k}^a} \;\in\; [-1, 1].$$

Order flow imbalance. The best-level event-time OFI of Cont, Kukanov, and Stoikov [6] is built from changes in displayed depth at the inside,

$$e_t \;=\; \mathbb{1}\{p_t^b \geq p_{t-1}^b\} q_t^b - \mathbb{1}\{p_t^b \leq p_{t-1}^b\} q_{t-1}^b - \mathbb{1}\{p_t^a \leq p_{t-1}^a\} q_t^a + \mathbb{1}\{p_t^a \geq p_{t-1}^a\} q_{t-1}^a,$$

and the aggregated OFI over a window is $\text{OFI}_{[\tau_1,\tau_2]} = \sum_{t:\tau_1 \leq t < \tau_2} e_t$.

VPIN. The volume-synchronized probability of informed trading of Easley, López de Prado, and O'Hara [7] groups trades into equal-volume buckets of size V and computes, within bucket j,

$$\text{VPIN}_j \;=\; \frac{1}{n} \sum_{i=j-n+1}^{j} \frac{|V_i^B - V_i^S|}{V},$$

with V_i^B and V_i^S the buy- and sell-classified volume in bucket i and n the window length in buckets.

Permanent vs transient impact. Hasbrouck [14] decomposes the mid-quote response to a trade into a permanent component (information) that persists in the efficient price and a transient component (liquidity) that decays as inventory works off.

A.10 Block bootstrap

The i.i.d. bootstrap is invalid for autocorrelated series because it destroys the serial structure that drives sampling uncertainty. The block bootstrap of Künsch [17] preserves it. From a series $\{z_t\}_{t=1}^T$, draw $\lceil T/L \rceil$ blocks of length L with replacement, where each block is a contiguous slice $(z_s, z_{s+1}, \ldots, z_{s+L-1})$ with s uniform on $\{1, \ldots, T\}$ with indices taken modulo T (the circular block bootstrap we use), concatenate them, and truncate to length T. Resample B times to form the bootstrap distribution.

The block length L governs the bias-variance trade-off: L must be large enough that adjacent blocks are approximately independent, but small enough that enough distinct blocks exist. We use $L = 300$ for hourly series, roughly twelve trading days.